

Open-Label, Uncontrolled, Multicenter Phase II Study to Evaluate the Efficacy and Toxicity of Cetuximab As a Single Agent in Patients With Recurrent and/or Metastatic Squamous Cell Carcinoma of the Head and Neck Who Failed to Respond to Platinum-Based Therapy

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A B S T R A C T

Purpose

To evaluate the efficacy and safety of the epidermal growth factor receptor–directed monoclonal antibody cetuximab administered as a single agent in patients with recurrent and/or metastatic squamous cell carcinoma of the head and neck (SCCHN) who experience disease progression on platinum therapy.

Patients and Methods

An open-label multicenter study in which patients with disease progression on two to six cycles of platinum therapy received single-agent cetuximab (initial dose 400 mg/m² followed by subsequent weekly doses of 250 mg/m²) for ≥ 6 weeks (single-agent phase). Patients who experienced disease progression could receive salvage therapy with cetuximab plus platinum (combination-therapy phase). From June 2001 to December 2002, 103 patients were enrolled and treated with cetuximab, 53 of whom subsequently received combination therapy.

Results

In the single-agent phase, response rate was 13%, disease control rate (complete response/partial response/stable disease) was 46%, and median time to progression (TTP) was 70 days. During the combination-therapy phase, the objective response rate was zero, disease control rate was 26%, and TTP was 50 days. Median overall survival was 178 days. Treatment was well tolerated. The most common cetuximab-related adverse events in the single-agent phase were skin reactions, particularly rash (49% of patients, mainly grade 1 or 2). There was one treatment-related death due to an infusion-related reaction.

Conclusion

Single-agent cetuximab was active and generally well tolerated in the treatment of recurrent and/or metastatic SCCHN that progressed on platinum therapy. Response was comparable to that seen with cetuximab plus platinum combination regimens in the same setting.

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INTRODUCTION

Head and neck cancers account for 6% of all cancers worldwide, with nearly 150,000 new cases in Europe alone each year.¹ For patients with recurrent and/or metastatic disease, the prognosis is extremely poor. For untreated head and neck cancers, the median survival is about 4 months.² Although response rates of 30% to 40% have been achieved consistently with cisplatin-based combination regimens, survival estimates remain at 6 to 9 months.³ Patients experiencing disease progression on platinum-based palliative chemotherapy have a

bleak prognosis, with a median survival of just more than 100 days.⁴

Among the novel therapies under development are those that target the human epidermal receptor (HER or erbB) superfamily. Of particular interest is HER1 (epidermal growth factor receptor [EGFR]).^{5,6} Studies in squamous cell carcinoma of the head and neck (SCCHN) have shown that 80% to 100% of tumors have an abnormal level of EGFR expression.⁶ There seems to be a positive correlation between levels of EGFR expression and poor prognosis, advanced disease, and reduced survival.⁷⁻⁹

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Cetuximab is an immunoglobulin G1 monoclonal antibody, designed specifically to block human EGFR.¹⁰ Cetuximab prolongs overall survival of locally advanced SCCHN patients when delivered in combination with radiation.^{11,12} It also enhances the antitumor activity of cisplatin in xenografts¹³ and has shown activity in the first-line treatment of recurrent and/or metastatic SCCHN in combination with cisplatin and cisplatin and fluorouracil.^{14,15} Recently, cetuximab with carboplatin/cisplatin has shown activity in platinum-resistant disease.^{16,17} However, the design of these studies meant it was not possible to assess whether the activity was due to reversal of platinum resistance by cetuximab and/or the intrinsic activity of cetuximab itself. Preliminary data from phase I studies with cetuximab alone have shown stable disease in patients with advanced tumors expressing high levels of EGFR.¹⁸ This study was designed to determine the efficacy and safety of single-agent cetuximab in patients with recurrent and/or metastatic SCCHN who had experienced treatment failure on platinum-based chemotherapy.

PATIENTS AND METHODS

Study Design and Treatment

This was an open-label, uncontrolled, multicenter phase II study, conducted at 19 centers in seven countries. The study had a two-phase design. In the first phase, all patients were assigned to receive cetuximab as an intravenous infusion at an initial dose of 400 mg/m² during 120 minutes, including a test dose of 20 mg, followed by weekly 1-hour infusions of 250 mg/m². Patients continued with cetuximab single-agent therapy for at least 6 weeks, if possible. If they responded to treatment or had stable disease (SD), treatment was continued until progressive disease (PD), clinical deterioration, or unacceptable adverse events were observed. Patients experiencing progression at this stage, as evidenced either by symptomatic deterioration and/or by means of imaging, were offered salvage therapy with cetuximab plus platinum. Patients continued to receive combination therapy until the occurrence of PD or unacceptable adverse effects. The study protocol and amendments were approved by independent ethics committees in each country. The study was conducted in accordance with the Declaration of Helsinki (1996); all patients provided written informed consent.

Patient Selection

Eligibility required histologically confirmed stage III/IV metastatic or recurrent SCCHN (according to American Joint Committee on Cancer) that was not suitable for local therapy, with documented PD within 30 days after a minimum of two and a maximum of six cycles of cisplatin-based (≥ 60 mg/m²/cycle) or carboplatin-based (≥ 300 mg/m²/cycle, or area under the time-concentration curve ≥ 4) chemotherapy. Additional patient eligibility criteria included age ≥ 18 years; Karnofsky performance status (KPS) $\geq 60\%$; measurable disease; tumor tissue available for immunohistochemical (IHC) staining of EGFR expression; and adequate hematologic, renal, and hepatic function. Exclusion criteria included nasopharyngeal carcinoma; concomitant malignant disease, except adequately treated basal cell cancer of the skin or cervical cancer in situ; chemotherapy (other than platinum based) or radiotherapy within the last 3 weeks; and prior or concomitant surgery within 30 days of study entry.

End Points and Statistics

The primary end point was the best overall response rate (complete responses [CRs] and partial responses [PRs]) with cetuximab single-agent therapy. The presence of PD at baseline and tumor responses were assessed by a blinded Independent Review Committee (IRC) comprising three radiologists and one oncologist.

Baseline evaluation of lesions was determined according to modified WHO criteria by computed tomography or magnetic resonance imaging. Response was evaluated every 6 weeks during single-agent therapy and at the

end of every other cycle (ie, every other third or fourth week depending on the platinum regimen) under combination therapy, starting in cycle 2. In the case of the combination therapy phase, the last assessment obtained before the therapy change was taken as baseline. Best response was based on assessments for index (maximum of 10) and nonindex lesions according to the following definitions: CR, disappearance of all index lesions (for nonindex lesions, no new lesions); PR, $\geq 50\%$ reduction in the sum of the products of diameters (SOPD) of index lesions compared with baseline SOPD, with no evidence of PD; SD, no sufficient decrease or increase in index lesions to qualify for PR or PD, respectively; PD, $\geq 25\%$ increase in the SOPD of index lesions compared with the smallest SOPD recorded for the study period (global nadir SOPD), or the appearance of one or more new lesions and/or unequivocal progression of existing nonindex lesions. CR or PR required confirmation after a minimum of 4 weeks.

Secondary end points for single-agent and combination-therapy phases were disease control rate (best response of either CR, PR, or SD), time to and duration of response (single-agent phase only), duration of response, time to progression (TTP, defined as the earlier of the number of days from the first dose of cetuximab to the first day of IRC-determined progression, or the day of death as a result of any cause within 60 days after the last tumor assessment), overall survival (OS; calculated as the number of days from the first dose of cetuximab until death, regardless of the cause; OS was evaluated during the two phases of the study), and changes in KPS from baseline.

Statistical Analyses

Continuous variables were summarized using descriptive statistics. Qualitative variables were summarized using counts and percentages. Two-sided CIs according to Clopper-Pearson¹⁹ were calculated for best overall response, and response and disease control rates. Kaplan-Meier estimates²⁰ were used for time-to-event end points. The primary population for efficacy analyses was the intention-to-treat (ITT) population (defined as all patients enrolled onto the study who received cetuximab), with supportive analyses of efficacy performed on the IRC-PD population (defined as patients in the ITT population who had documented PD, as determined by the IRC, within 30 days after the end of the last cycle of previous platinum-based chemotherapy). Safety analyses were conducted on the ITT population. Statistical analyses were carried out using the Statistical Analysis System software, version 6.12 (SAS Institute, Cary, NC). All findings up to the cutoff dates of August 4, 2003, (for clinical findings) and December 31, 2003, (for survival) were included in the analyses.

Pretreatment and On-Study Evaluations

EGFR expression was evaluated centrally in tumor tissue taken during surgery or at biopsy before the start of the study using a standardized IHC assay (DakoCytomation, Glostrup, Denmark). The percentage of stained cells and the staining intensity (0, nondetectable; 1+, faint; 2+, weak; 3+, strong) were evaluated.

Blood samples were taken on a weekly basis before the cetuximab infusion. Blood samples for the determination of cetuximab serum concentrations (2.5 mL per sample) were taken just before the start of the infusion and at the end of the infusion at visits 1, 4, and 6 for single-agent and combination therapy. A sample also was taken at the end-of-study visit. A validated sandwich enzyme-linked immunosorbent assay was used to determine serum concentrations of cetuximab.

Adverse events were recorded weekly in both phases and graded according to National Cancer Institute Common Toxicity Criteria version 2 and described using the Coding Symbols for a Thesaurus of Adverse Reaction Terms (1995) dictionary.

RESULTS

A total of 103 patients were enrolled onto the study between June 2001 and December 2002. All were included in the ITT population. Of these, 53 patients (51%) who experienced progression while receiving single-agent cetuximab continued treatment with

combination therapy (cetuximab plus either cisplatin or carboplatin). The IRC-PD population (ie, those patients with IRC-determined PD at study entry) included 66 patients. The difference in the numbers of patients in the ITT and IRC-PD populations was due in all cases to discrepancies between the investigators' and the IRC's interpretation of PD at study entry. On December 31, 2003, eight patients were still undergoing treatment.

Patient Characteristics

Demographic and clinical characteristics for the ITT and IRC-PD populations are listed in Table 1. The ITT population comprised 84 males and 19 females (median age, 57 years). The most common site of the primary tumor was the pharynx (38%; Table 2). The disease was metastatic in 80% of patients (located at distant sites in 48%). EGFR expression was determined in 100 of 103 (97%) patients and EGFR-expressing cells were found in 97% (97 of 100) of these patients (94% overall).

Cisplatin and carboplatin alone had been administered in seven (7%) and 16 patients (16%), respectively. The remaining 80 patients (78%) had received cisplatin and/or carboplatin in combination with other agents. The median time from the end of platinum-based chemotherapy to the first dose of cetuximab in the ITT population was 15 days (range, 0 to 189 days).

Exposure to Cetuximab

During the total study period, 103 patients received a median of 15 cetuximab infusions (range, one to 53 infusions) for a median of 14.1 weeks (range, 0.1 to 56.2 weeks). Seventy-two patients (70%) received between six and 30 infusions; 14 patients (14%) received between 30 and 50 infusions, and one patient had received more than 50 infusions by the cutoff date.

Response and Disease Control Rates

During the single-agent phase, the response rates were similar for the ITT and IRC-PD populations (Table 3). In the ITT population, response was not assessable in 18 patients (17%) because the scans

Characteristic	ITT Population (n = 103)	IRC-PD Population (n = 66)
Male		
No. of patients	84	50
%	82	76
Female		
No. of patients	19	16
%	18	24
Age, years		
Median	57	56
Range	23-77	23-73
KPS, %		
Median	80	80
Range	60-100	60-100
Disease duration, months		
Median	20.1	19.1
Range	2.5-282.9	3-163.3

Abbreviations: ITT, intention to treat; IRC-PD, ITT patients with progressive disease as determined by the Independent Review Committee; KPS, Karnofsky performance status.

Table 2. Baseline Characteristics of the ITT Population (n = 103)

Characteristic	No. of Patients	%
EGFR expression*	97†	97
Primary tumor at screening		
Pharynx	39	38
Larynx	21	20
Paranasal sinuses	3	3
Other	34	33
Nonclassifiable	6	6
Tumor grade at primary diagnosis		
Well differentiated	24	23
Moderately differentiated	34	33
Poorly differentiated/undifferentiated	19	18
Unknown/missing†	26	25
Previous platinum-based chemotherapy		
Cisplatin alone	7	7
Carboplatin alone	16	16
In combination with other agent(s)	70	68
Cisplatin + carboplatin + FU	10	10
Duration of previous platinum-based chemotherapy to documentation of PD, days		
Median	66	
Range	20-162	
Other prior SCCHN therapy	85	83
Chemotherapy (platinum)	20	19
Chemotherapy (nonplatinum)	50	49
Radiotherapy	81	79
Surgery	76	74

Abbreviations: ITT, intention to treat; EGFR, epidermal growth factor receptor; FU, fluorouracil; PD, progressive disease; SCCHN, squamous cell carcinoma of the head and neck.

*n = 100; EGFR was not determined in three patients.

†Missing for one patient.

were either poor quality or missing. The best overall response and disease control rates were 13% (95% CI, 7% to 21%) and 46% (95% CI, 36% to 56%), respectively. No patient achieved a best response of CR in either population. The median time to response and duration of response was 49 days (range, 37 to 251 days) and 126 days, respectively.

During the combination phase, the response rates were also similar for the ITT and IRC-PD populations. In the ITT population, response was not assessable in 25 patients (47%) because the scans were either poor quality or missing. The objective response rate was zero and the disease control rate was 26% (Table 3).

TTP and OS

In the ITT population, the median TTP was 70 days in the single-agent phase and 50 days in the combination-therapy phase (Fig 1). By December 31, 2003, 93 (90%) of the 103 patients in the ITT population had died; the median survival time was 178 days (Fig 2).

KPS Trends and Prognostic Factors

Although baseline KPS was very good and even excellent in some patients, as expected, scores worsened as the study progressed. However, the best status achieved was an improvement in 42 patients (45%), which complies with the disease control rate of 46% observed in the ITT population. A similar pattern was observed in the IRC-PD population. There appeared to be a clear relationship between baseline KPS and a prolonged OS (Table 4), and a relationship between response and degree of change in KPS. Given that KPS was no longer

Table 3. Summary of Best Response and Disease Control Rate in Single-Agent and Combination-Therapy Phases

Response Variable	Single-Agent Phase				Combination-Therapy Phase			
	ITT Population (n = 103)		IRC-PD Population (n = 66)		ITT Population (n = 53)		IRC-PD Population (n = 37)	
	No.	%	No.	%	No.	%	No.	%
Best response								
Complete response*	0		0		0		0	
Partial response*	13	13	8	12	0		0	
Stable disease	34	33	22	33	14	26	10	27
Progressive disease	38	37	23	35	14	26	9	24
Image not assessable	16	16	12	18	5	9	3	8
No image available	2	2	1	2	20	38	15	41
Best overall response rate		13		12		0		0
95% CI		7 to 21		5 to 22		0 to 7		0 to 10
Disease control rate		46		45		26		27
95% CI		36 to 56		33 to 58		15 to 40		14 to 44

Abbreviations: ITT, intention to treat; IRC-PD, ITT patients with progressive disease as determined by the Independent Review Committee.

*Confirmed responses (ie, those persisting for at least 4 weeks).

recorded once PD was observed, the number of patients with KPS data declined rapidly, which limits the interpretation of these findings.

In subgroup analyses of baseline characteristics, only the absence of metastases was associated with an improvement in all outcome variables (Tables 4 and 5). No relationship was observed between efficacy and the degree of baseline EGFR expression.

Early Skin Reactions

During the single-agent phase, most episodes of skin reaction and acne-like rash developed within the first 3 weeks of treatment; none of which were grade 3 or 4. The early development of grade 1 or 2 skin reactions after treatment with single-agent therapy was not associated with prolonged TTP or OS (Table 5), nor was there any relationship between early-onset skin reaction or acne-like rash and response or disease control rates. However, there was a trend toward higher response and disease control rates in patients who developed any skin reaction or an acne-like rash compared with those not developing skin reactions (Table 5).

Safety and Tolerability

Adverse events were reported for 102 patients (99%) treated with single-agent therapy and 52 patients (98%) treated with combination therapy. The most common or relevant cetuximab-related adverse events in both the single-agent and combination-therapy phase were rash, acne, and asthenia (Table 6). Adverse events were generally mild to moderate (grades 1 and 2); grade 3/4 events were observed in $\leq 6\%$ of patients for each category. Six patients had infusion-related reactions during the single-agent phase; all episodes were considered related to cetuximab treatment.

Serious adverse events, mostly grade 3 or 4, were reported in 47 patients (46%) in the single-agent phase and 26 patients (49%) in the combination phase; they were considered cetuximab related in 21 patients (20%) and three patients (6%), respectively. There was one cetuximab-related death. This patient suffered a fatal infusion-related reaction to the first infusion of cetuximab.

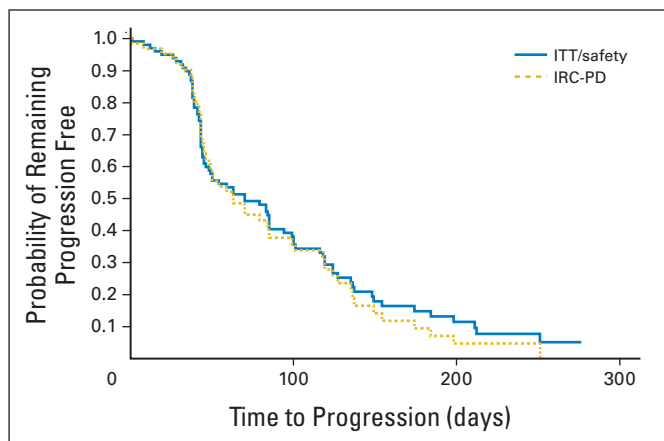


Fig 1. Kaplan-Meier estimates for the time to progression (days) from the start of cetuximab treatment in the intention-to-treat (ITT)/safety population and patients in the ITT population who had documented progressive disease, as determined by the Independent Review Committee (IRC-PD; single-agent therapy phase).

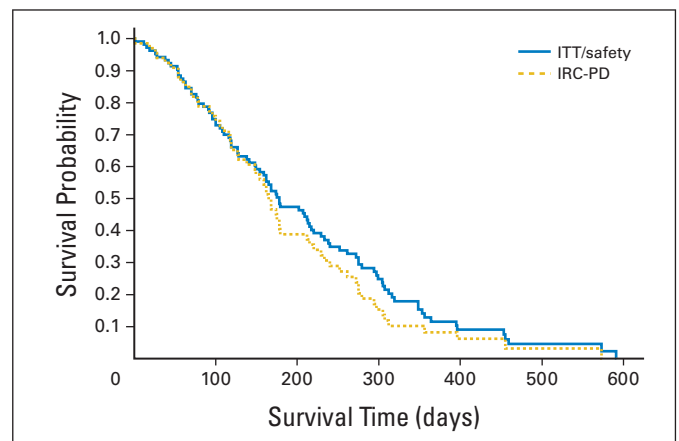


Fig 2. Kaplan-Meier estimates for survival time (days) from the start of cetuximab treatment in the intention-to-treat (ITT)/safety population and patients in the ITT population who had documented progressive disease, as determined by the Independent Review Committee (IRC-PD).

Table 4. Subgroup Analyses According to Baseline Demographic and Disease Characteristics (ITT population)

Subgroup	No. of Patients	Single-Agent Phase			Both Phases: Median OS (days)
		Response Rate (%)	Disease Control Rate (%)	Median TTP (days)	
All patients	103	13	46	70	178
Age, years					
< 65	79	13	44	70	174
≥ 65	24	13	50	83	193.5
EGFR expression					
0	3	0	33	41	85
1+	1	0	100	101	165
2+	7	14	29	43	119
3+	89	14	47	83	209
KPS, %					
< 80	38	13	47	70	138
≥ 80	64	13	45	84	202
AJCC stage					
III	9	11	56	63	159
IV	94	13	45	70	178
Distant metastases					
Yes	49	8	33	45	162
No	54	17	57	94	178

Abbreviations: ITT, intention to treat; TTP, time to progression; OS, overall survival; EGFR, epidermal growth factor receptor; KPS, Karnofsky performance status; AJCC, American Joint Committee on Cancer.

Pharmacokinetic Results

Exposure to cetuximab was similar after cetuximab monotherapy and platinum-based chemotherapy (data not shown).

DISCUSSION

In this phase II study, strict inclusion criteria ensured that the patient population was a homogenous, clearly platinum-refractory

group of SCCHN patients with a poor prognosis. The study population's characteristics were typical of this patient group, with frequent comorbidities and a history of extensive previous therapy.

Single-agent cetuximab showed encouraging clinical efficacy. Although there were discrepancies between the investigators' and the IRC's assessment of progressive disease (due mainly to the stringent criteria used by the IRC), which led to a difference in the ITT and IRC-PD population sizes, the overall response and disease control rates were similar in both populations. The median TTP

Table 5. Response Rate, Disease Control Rate, TTP, and OS According to the Occurrence of Skin Reactions During the Single-Agent Therapy Phase (ITT population)

Characteristic	No. of Patients	Response Rate (n = 103)	Disease Control Rate (n = 103)	No. of Patients	Median TTP (days)* (n = 92)	No. of Patients	Median OS (days)* (n = 99)
All patients	103	13%	46%	92†	62	99‡	158
Early skin reactions§							
None	41	15%	44%	35	64	38	153
Grades 1/2	62	11%	47%	57	49	61	158
Early acne-like rash§							
None	51	14%	49%	45	64	48	181
Grades 1/2	52	12%	42%	47	49	51	157
Skin reactions							
None	24	4%	25%	—	—	—	—
Grades 1/2	78	15%	51%	—	—	—	—
Grades 3/4	1	0	100%	—	—	—	—
Acne-like rash							
None	28	7%	25%	—	—	—	—
Grades 1/2	74	15%	53%	—	—	—	—
Grades 3/4	1	0	100%	—	—	—	—

Abbreviations: TTP, time to progression; OS, overall survival; ITT, intention to treat; COSTART, Coding Symbols for a Thesaurus of Adverse Reaction Terms (1995) dictionary.

*Time from 3 weeks after the first dose of cetuximab to progression or death.

†Eleven patients dropped out of the study or died by end of the third week of single-agent therapy.

‡Four patients had died by the end of the third week of single-agent therapy.

§Within the first 3 weeks of therapy. Skin reactions are described according to COSTART codes.

Table 6. Frequencies of Most Common or Relevant Cetuximab-Related AEs by Grade of National Cancer Institute–Common Toxicity Criteria (ITT population)

Special AE Category	AE Grade	Single-Agent Therapy Phase (n = 103)		Combination-Therapy Phase (n = 53)	
		No. of Patients	%	No. of Patients	%
Rash	All grades	50	49	10	19
	3 or 4	1	1	1	2
Acne	All grades	27	26	3	6
	3 or 4	0	0	0	0
Asthenia	All grades	25	24	6	11
	3 or 4	4	4	3	6
Nail disorder	All	16	16	2	4
	3 or 4	0	0	1	2
Dry skin	All	14	14	3	6
	3 or 4	0	0	0	0
Fever	All	14	14	1	2
	3 or 4	1	1	1	2
Nausea	All	13	13	10	19
	3 or 4	1	1	0	0
Vomiting	All	11	11	4	8
	3 or 4	2	2	0	0
Dyspnea	All	5	5	0	0
	3 or 4	4	4	0	0
Infusion-related reactions	All	6	6	0	0
	3 or 4	1	1	0	0

NOTE: If a patient experienced more than one adverse event, the patient was counted once according to the highest toxicity grade in that category. Abbreviations: AE, adverse event; ITT, intention to treat.

and median OS in the ITT population were 70 days and 178 days, respectively. However, once the patient had experienced PD while receiving cetuximab alone, the efficacy of cetuximab in combination with platinum was marginal, with no objective response, although 26% of the patients in the ITT population showed SD. These preliminary efficacy results represent a considerable improvement over current second-line chemotherapies available for patients experiencing PD while receiving platinum therapy and may provide a survival benefit.

There are no published findings with which to compare the results of this study directly with single-agent cetuximab. However, our results closely mirror those shown in two phase II studies with cetuximab in combination with platinum in similar populations of patients with PD while receiving platinum.^{16,17} Overall response rates were 10%,^{16,17} with corresponding disease control rates of 56% and 53%. Baselga et al¹⁷ reported a median TTP and OS of 85 days and 183 days, respectively, which were increased to 204 days and 294 days, respectively, in responders. Together with our observation that the combination of platinum and cetuximab in patients who experienced PD while receiving cetuximab monotherapy served only to stabilize disease in a proportion of patients but did not induce a clinical response, this raises questions about the clinical usefulness of platinum in combination with cetuximab in patients who experienced PD while receiving platinum. Interestingly, the results we report are in contrast to the situation reported in metastatic colorectal cancer, in which the combination of irinotecan and cetuximab led to higher response rates and longer median survival than cetuximab monotherapy in patients who experienced treatment failure after irinotecan therapy.²¹

Cetuximab monotherapy was well tolerated. One patient died as a result of an infusion-related reaction after cetuximab treatment. There was no change in the safety profile when cetuximab was administered in combination with cisplatin or carboplatin. No unexpected adverse events were observed and all were either consistent with the underlying disease, including respiratory disorders, or the known safety profile of cetuximab or platinum drugs. Rash is commonly seen with EGFR-targeted agents.²² In both phases of this study most skin reactions were grade 1 or 2.

We are searching constantly for factors that will help us to direct therapy to those patients who will most benefit from it. After cetuximab treatment, a relationship between rash and efficacy has been documented in SCCHN^{14,16} and other malignancies.²³ It also has been seen with tyrosine kinase inhibitors.^{24,25} In contrast, we observed no link between skin reactions and outcome. The severity of the rash may be a predictor of survival advantage,²³ and the fact that we observed so few episodes of grade 3 or 4 rash might explain, in part, the lack of a link between rash and outcome in our study.

Baseline KPS was not associated with response or TTP in this population, but did have a clear link with OS.

There were too few data in this study to allow speculation on the relationship between EGFR expression and response or survival. Findings from a retrospective analysis and clinical studies in metastatic colorectal cancer indicated that EGFR levels detected by IHC do not seem necessary for a response to treatment.^{26,27} The utility of IHC for detecting EGFR expression, to select patients most likely to benefit from treatment with EGFR inhibitors, is being increasingly called into question. Molecular profiling of the tumor, in terms of gene mutation (point or amplification) and/or expression, may provide the key to patient selection. EGFR tyrosine kinase mutations have only rarely been reported in SCCHN,²⁸ and correlation with response to tyrosine kinase inhibitors is unclear.²⁹ The constitutively active EGFR variant, EGFRvIII, is expressed in SCCHN and may affect tumor response to cetuximab,³⁰ but there are no clinical data yet to support this. EGFR gene mutations predicting response to cetuximab have not yet been identified.

Cetuximab has also demonstrated activity in the treatment of recurrent and/or metastatic SCCHN in the first-line setting^{14,15,31} and in locally advanced disease.¹¹

Thus, there is a clear line of evidence pointing to cetuximab as an important future treatment option in SCCHN. However, the most effective way of using cetuximab at the various stages of recurrent and/or metastatic SCCHN still has to be defined. Our preliminary data provide evidence that in patients experiencing PD while receiving platinum therapy, single-agent cetuximab is effective and comparable to combination therapy with cetuximab and platinum-based regimens in the same setting.

AUTHORS' DISCLOSURES OF POTENTIAL CONFLICTS OF INTEREST

Although all authors completed the disclosure declaration, the following authors or their immediate family members indicated a financial interest. No conflict exists for drugs or devices used in a study if they are not being evaluated as part of the investigation. For a detailed description of the disclosure categories, or for more information about ASCO's conflict of interest policy, please refer to the Author Disclosure Declaration and the Disclosures of Potential Conflicts of Interest section in Information for Contributors.

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REFERENCES

1. International Agency for Research on Cancer: CANCERmondial. www-dep.iarc.fr
2. Kowalski LP, Carvalho AL: Natural history of untreated head and neck cancer. *Eur J Cancer* 36:1032-1037, 2000
3. Cohen EE, Lingen MW, Vokes EE: The expanding role of systemic therapy in head and neck cancer. *J Clin Oncol* 22:1743-1752, 2004
4. León X, Hitt R, Constenla M, et al: A retrospective analysis of the outcome of patients with recurrent and/or metastatic squamous cell carcinoma of the head and neck refractory to a platinum-based chemotherapy. *Clin Oncol (R Coll Radiol)* 17:418-424, 2005
5. Salomon DS, Brandt R, Ciardiello F, et al: Epidermal growth factor-related peptides and their receptors in human malignancies. *Crit Rev Oncol Hematol* 19:183-232, 1995
6. Herbst RS, Langer CJ: Epidermal growth factor receptors as a target for cancer treatment: The emerging role of IMC-C225 in the treatment of lung and head and neck cancers. *Semin Oncol* 29:27-36, 2002
7. Nicholson RI, Gee JM, Harper ME: EGFR and cancer prognosis. *Eur J Cancer* 37:S9-S15, 2001 (suppl 4)
8. Grandis JR, Melhem MF, Barnes EL, et al: Quantitative immunohistochemical analysis of transforming growth factor- α and epidermal growth factor receptor in patients with squamous cell carcinoma of the head and neck. *Cancer* 78:1284-1292, 1996
9. Ang KK, Berkey BA, Tu X, et al: Impact of epidermal growth factor receptor expression on survival and pattern of relapse in patients with advanced head and neck carcinoma. *Cancer Res* 62:7350-7356, 2002
10. Goldstein NI, Prewett M, Zuklys K, et al: Biological efficacy of a chimeric antibody to the epidermal growth factor receptor in a human tumor xenograft model. *Clin Cancer Res* 1:1311-1318, 1995
11. Bonner JA, Harari PM, Giralto J, et al: Radiotherapy plus cetuximab for squamous-cell carcinoma of the head and neck. *N Engl J Med* 354:567-578, 2006
12. Robert F, Ezekiel MP, Spencer SA, et al: Phase I study of anti-epidermal growth factor receptor antibody cetuximab in combination with radiation therapy in patients with advanced head and neck cancer. *J Clin Oncol* 19:3234-3243, 2001
13. Fan Z, Baselga J, Masui H, et al: Antitumor effect of anti-epidermal growth factor receptor monoclonal antibodies plus cis-diamminedichloroplatinum on well established A431 cell xenografts. *Cancer Res* 53:4637-4642, 1993
14. Burtneess B, Goldwasser MA, Flood W, et al: Phase III randomized trial of cisplatin plus placebo compared with cisplatin plus cetuximab in metastatic/recurrent head and neck cancer: An Eastern Cooperative Oncology Group study. *J Clin Oncol* 23:8646-8654, 2005
15. Bourhis J, Rivera F, Mesia R, et al: Phase I/II study of cetuximab in combination with cisplatin or carboplatin and fluorouracil in patients with recurrent or metastatic squamous cell carcinoma of the head and neck. *J Clin Oncol* 24:2866-2872, 2006
16. Herbst RS, Arquette MA, Shin D, et al: Epidermal growth factor receptor antibody cetuximab and cisplatin for recurrent and refractory squamous cell carcinoma of the head and neck: A phase II, multicenter study. *J Clin Oncol* 23:5578-5587, 2005
17. Baselga J, Trigo JM, Bourhis J, et al: Phase II multicenter study of the antiepidermal growth factor receptor monoclonal antibody cetuximab in combination with platinum-based chemotherapy in patients with platinum-refractory metastatic and/or recurrent squamous cell carcinoma of the head and neck. *J Clin Oncol* 23:5568-5577, 2005
18. Baselga J, Pfister D, Cooper MR, et al: Phase I studies of anti-epidermal growth factor receptor chimeric antibody C225 alone and in combination with cisplatin. *J Clin Oncol* 18:904-914, 2000
19. Clopper CJ, Pearson E: The use of confidence or fiducial limits illustrated in the case of binomial. *Biometrika* 26:404-413, 1934
20. Kaplan EL, Meier P: Nonparametric estimation from incomplete observations. *J Am Stat Assoc* 53:457-481, 1958
21. Cunningham D, Humblet Y, Siena S, et al: Cetuximab monotherapy and cetuximab plus irinotecan in irinotecan-refractory metastatic colorectal cancer. *N Engl J Med* 351:337-345, 2004
22. Perez-Soler R: Can rash associated with HER1/EGFR inhibition be used as a marker of treatment outcome? *Oncology (Williston Park)* 17:23-28, 2003 (suppl 12)
23. Saltz L, Kies M, Abbruzzese JL, et al: The presence and intensity of the cetuximab-induced acne-like rash predicts increased survival in studies across multiple malignancies. *Proc Am Soc Clin Oncol* 22:204 2003 (abstr 817)
24. Clark GM, Perez-Soler R, Siu L, et al: Rash severity is predictive of increased survival with erlotinib HCl. *Proc Am Soc Clin Oncol* 22:196 2003 (abstr 786)
25. Cohen EE, Rosen F, Stadler WM, et al: Phase II trial of ZD1839 in recurrent or metastatic squamous cell carcinoma of the head and neck. *J Clin Oncol* 21:1980-1987, 2003
26. Lenz HJ, Van Cutsem E, Khambata-Ford S, et al: Multicenter phase II and translational study of cetuximab in metastatic colorectal carcinoma refractory to irinotecan, oxaliplatin, and fluoropyrimidines. *J Clin Oncol* 24:4914-4921, 2006
27. Chung KY, Shia J, Kemeny NE, et al: Cetuximab shows activity in colorectal cancer patients with tumors that do not express the epidermal growth factor receptor by immunohistochemistry. *J Clin Oncol* 23:1803-1810, 2005
28. Willmore-Payne C, Holden JA, Layfield LJ: Detection of EGFR- and HER2-activating mutations in squamous cell carcinoma involving the head and neck. *Mod Pathol* 19:634-640, 2006
29. Cohen EE, Lingen MW, Martin LE, et al: Response of some head and neck cancers to epidermal growth factor receptor tyrosine kinase inhibitors may be linked to mutation of ERBB2 rather than EGFR. *Clin Cancer Res* 11:8105-8108, 2005
30. Sok JC, Coppelli FM, Thomas SM, et al: Mutant epidermal growth factor receptor (EGFRvIII) contributes to head and neck cancer growth and resistance to EGFR targeting. *Clin Cancer Res* 12:5064-5073, 2006
31. Vermorken J, Mesia R, Vega-Villegas ME, et al: Cetuximab in combination with cisplatin or carboplatin and 5-fluorouracil (FU) in the first-line treatment of patients with recurrent and/or metastatic squamous cell carcinoma of the head and neck (R&M SCCHN) (EXTREME). *J Clin Oncol* 24:289s, 2006 (suppl; abstr 5537)